NavBot Chunking and Retrieval Methodology

Chunking

Tried 64, 128, and 256 token windows with a 32-token overlap.
256 selected: best accuracy at fixed settings, average context near 1.1k,
and it fits the embedding model's 256-token input limit.

Chunk size	Chunks	Correct	Avg context
64	407	31/59	337
128	133	34/59	586
256	57	37/59	1,091

Rerank 4 vs 8 at 256 tokens: top 8 added one correct case but raised
average context from 1,091 to 2,174; top 4 was retained.

Chunk	Top 4	Top 8	Avg context 4 / 8
64	31/59	32/59	337 / 669
128	34/59	33/59	586 / 1,170
256	37/59	38/59	1,091 / 2,174

Retrieval

Vector store: ChromaDB, persistent cosine HNSW.
Dense: all-MiniLM-L6-v2, top 20.
BM25: in-memory from Chroma; used with dense retrieval.
Fusion: reciprocal rank fusion; combined both rankings.
Reranker: cross-encoder/ms-marco-MiniLM-L-6-v2; top 4.
BM25 + RRF did not change the score much.

Backend

Offline: DOCX -> chunks -> embeddings -> ChromaDB.
Online: request -> FastAPI -> NavBotChat -> dense + BM25 -> RRF ->
cross-encoder -> LLM -> answer + sources.

Runtime: ONNX Runtime on CPU; both models use 8-bit quantized weights.